Final Project Idea

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CUDA Memory Copy

Host ↔ Device

- `cudaMemcpyAsync*` family of functions
- Not the focus for this project, but pretty nice to have

Global → Shared

- `cuda::memcpy_async`
- Named producer-consumer (“Arrive/Wait”) barriers
- Comes with your choice of an ordering model
  - Acquire
  - Release
  - Relaxed

Final Project Idea

- A “DSL” to write fused kernels without data races
- Static checker for data races (and possibly equivalence)
- Procedural/statement-oriented language
- Annotate each place expression with ordering/barrier info

A place expression is equivalent to a `restricted` lvalue in C.

- Evaluation
  - sgemm (A×B, A×B+C×D)
  - SpMV
  - Other ideas…?